

Task2

Task2(a):

In this query we first find all the products in the “Books” category, get their reviews, calculate the average of the ratings. Then it counts the total number of times that book was purchased by a user. Finally, we filter the ratings that has the average between 3 and 4 “inclusive” and purchased more than 5 times, to return product_name, totalPurchases and the averageRating sorted from the highest to the lowest.

```
MATCH (p:Product)-[:BELONGS_TO]->(c:Category {name:
"Books"})
MATCH (r:Review)-[:REVIEWS]->(p)
WITH p, avg(r.rating) AS averageRating
MATCH (u:User)-[:PURCHASED]->(p)
WITH p, averageRating, count(u) AS totalPurchases
WHERE averageRating >= 3 AND averageRating <= 4 AND
totalPurchases > 5
RETURN p.product_name, averageRating, totalPurchases
ORDER BY averageRating DESC
```

```

1 MATCH (p:Product)-[:BELONGS_TO]->(c:Category {name: "Books"})
2 MATCH (r:Review)-[:REVIEWS]->(p)
3 WITH p, avg(r.rating) AS averageRating
4 MATCH (u:User)-[:PURCHASED]->(p)
5 WITH p, averageRating, count(u) AS totalPurchases
6 WHERE averageRating >= 3 AND averageRating <= 4 AND totalPurchases > 5
7 RETURN p.product_name, averageRating, totalPurchases
8 ORDER BY averageRating DESC

```

Table RAW

	p.product_name	averageRating	totalPurchases
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1	"Zenith Ago"	3.5	6
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2	"Harbor Necessary"	3.3333333333333335	6
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Started streaming 2 records after 1 ms and completed after 12 ms.

Task2(b):

In this query we link every order to the product it contains and count the total number of products per order. Then, we only filter those with more than 3 products. Finally, we return the order_id and the sorted productCount.

```

MATCH (o:Order)-[:CONTAINS]->(p:Product)
WITH o, count(p) AS productCount
WHERE productCount > 3
RETURN o.order_id, productCount
ORDER BY productCount DESC

```

```
1 MATCH (o:Order)-[:CONTAINS]->(p:Product)
2 WITH o, count(p) AS productCount
3 WHERE productCount > 3
4 RETURN o.order_id, productCount
5 ORDER BY productCount DESC
```

	o.order_id	productCount
1	"000004803"	6
2	"000018821"	6
3	"000001651"	6
4	"000012257"	6
5	"000015872"	6
6	"000012881"	6
7	"000017178"	6
8	"000012252"	6
9	"000009509"	6
10	"000002415"	6

Started streaming 3,072 records after 11 ms and completed after 90 ms.

Task2(c):

In this query we get all users who placed that contains the product "Harbor Two", then for those same users we get the other products placed by them after removing the duplicates.

```
MATCH (target:Product {product_name: "Harbor Two"})<-
[:CONTAINS]-(:Order)<-[:PLACED]-(u:User)
MATCH (u)-[:PLACED]->(:Order)-[:CONTAINS]->(other:Product)
WHERE other.product_name <> "Harbor Two"
```

RETURN DISTINCT other.product_name

```
1 MATCH (target:Product {product_name: "Harbor Two"})<-[:CONTAINS]-(:Order)<-[:PLACED]-(u:User)
2 MATCH (u)-[:PLACED]->(:Order)-[:CONTAINS]->(other:Product)
3 WHERE other.product_name <> "Harbor Two"
4 RETURN DISTINCT other.product_name
```

Table RAW

other.product_name

1	"Acme Two"
2	"Harbor Eye"
3	"Acme Show"
4	"Astra Tough"
5	"Orion Make"
6	"Nimbus Significant"
7	"Acme Black"
8	"Pulse Certainly"
9	"GreenLeaf Improve"
10	"GreenLeaf Bill"

Started streaming 173 records after 3 ms and completed after 7 ms.

Task2(d):

This query matches categories with products, and filter for category name to be between 'C' and 'H', then it returns the category name, the number of products in it and a list of all those products names.

```
MATCH (c:Category)<-[:BELONGS_TO]-(p:Product)
WHERE substring(c.name, 0, 1) >= 'C' AND substring(c.name, 0, 1)
<= 'H'
```

```
RETURN c.name AS Category Name ,
count(p) AS Number of Products ,
collect(p.product_name) AS Products List
```

```
1 MATCH (c:Category)-[:BELONGS_TO]-(p:Product)
2 WHERE substring(c.name, 0, 1) >= 'C' AND substring(c.name, 0, 1) <= 'H'
3 RETURN c.name AS `Category Name`,
4       count(p) AS `Number of Products`,
5       collect(p.product_name) AS `Products List`
```

Category Name	Number of Products	Products List
"Clothing"	213	["Astra Be", "Everest Friend", "Nimbus Sit", "Willow Budget", "Orion Itself", "NeoTech A", "Acme Significant", "Everest Easy", "Orion Short", "Acme Fire", "Willow Approach", "Everest As", "NeoTech Certainly", "Willow Cultural", "Everest Interesting", "Zenith Coach", "GreenLeaf Bar", "Harbor Until", "Pulse Order", "Zenith White", "Solace Everyone", "Everest Smile", "NeoTech Onto", "Willow Recently", "Zenith Way", "Pulse Kitchen", "Orion Most", "Harbor Itself", "Willow Study", "GreenLeaf Painting", "Zenith Within", "Zenith Follow", "Solace Six", "Solace School", "Astra Well", "Orion Idea", "Willow Site", "Orion Side", "Zenith Religious", "Acme Outside", "Harbor Man", "GreenLeaf Discover", "Solace Change", "Orion Film", "Everest F

Started streaming 4 records after 1 ms and completed after 5 ms.

Task2(e):

In this query we create first the user node with my information. Then select 4 different products from the database. Then it creates relationships to show the user viewed and added the items to a cart. After that we create a new order node linked to the created user, then with unwind we connect the order to each item. Finally, it records the

purchase of each product, generates 5 stars review for each one, and return the user name, order_id and a list of purchased products.

```
CREATE (me:User {name: "Az-El-Arab Chakour", gender: "Male",
city: "Clausthal-Zellerfeld", user_id: "AZ-2026", email: "az-el-
arab.chakour@tu-clausthal.de"})
WITH me
MATCH (p:Product)
WHERE p.product_name IN ["Astra Small", "Everest Push", "Orion
Society", "Nimbus Or"]
WITH me, p
CREATE (me)-[:VIEWED]->(p)
CREATE (me)-[:ADDED_TO_CART]->(p)
WITH me, collect(p) AS cart
CREATE (o:Order {order_id: "ORD-AZ-99", order_date: "2026-05-
26", total_amount: 150.0})
CREATE (me)-[:PLACED]->(o)
WITH me, o, cart
UNWIND cart AS p
CREATE (o)-[:CONTAINS]->(p)
CREATE (me)-[:PURCHASED]->(p)
CREATE (r:Review {review_id: "REV-AZ-" + p.product_id, rating:
5})
CREATE (me)-[:WROTE]->(r)
CREATE (r)-[:REVIEWS]->(p)
RETURN me.name, o.order_id, collect(p.product_name) AS
```

purchasedProducts

```
1 CREATE (me:User {name: "Az-El-Arab Chakour", gender: "Male",
2   city: "Clausthal-Zellerfeld", user_id: "AZ-2026", email: "az-
3   el-arab.chakour@tu-clausthal.de"})
4 WITH me
5 MATCH (p:Product)
6 WHERE p.product_name IN ["Astra Small", "Everest Push",
7   "Orion Society", "Nimbus Or"]
8 WITH me, p
9 CREATE (me)-[:VIEWED]->(p)
10 CREATE (me)-[:ADDED_TO_CART]->(p)
11 WITH me, collect(p) AS cart
12 CREATE (o:Order {order_id: "ORD-AZ-99", order_date: "2026-05-
13   26", total_amount: 150.0})
14 CREATE (me)-[:PLACED]->(o)
15 WITH me, o, cart
16 UNWIND cart AS p
17 CREATE (o)-[:CONTAINS]->(p)
18
19 CREATE (me)-[:PURCHASED]->(p)
20 CREATE (r:Review {review_id: "REV-AZ-" + p.product_id,
21   rating: 5})
22 CREATE (me)-[:WROTE]->(r)
23 CREATE (r)-[:REVIEWS]->(p)
24 RETURN me.name, o.order_id, collect(p.product_name) AS
   purchasedProducts
```



No graph data available
Your query returned records, but they
can't be visualized as a graph.

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me.name	o.order_id	purchasedProducts
"Az-El-Arab Chakour"	"ORD-AZ-99"	["Astra Small", "Everest Push", "Nimbus Or", "Orion Society"]